Foundations of Data Science

Lecture 1: Build, compute, critique, repeat

Prof. Gilles Louppe
g.louppe@uliege.be

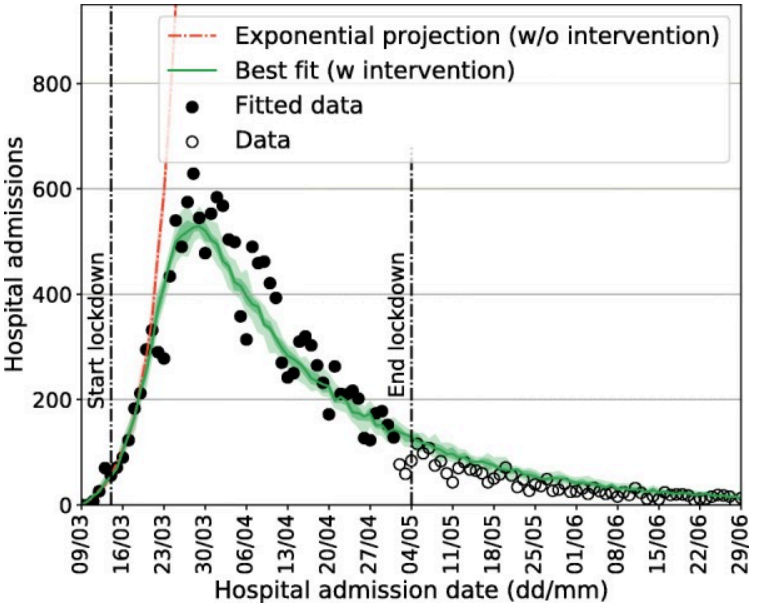
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Three stories

Before any theory, three stories from the last few years.

Each one follows the same four steps: **build** a model of how the data came to be, **compute** what the data say, **critique** the answer, and **repeat**.

COVID-19 in Belgium



① **Build**: a model of how the virus spreads, by age group, contacts and mobility.

② **Compute**: fit to daily admissions, then predict with and without lockdown.

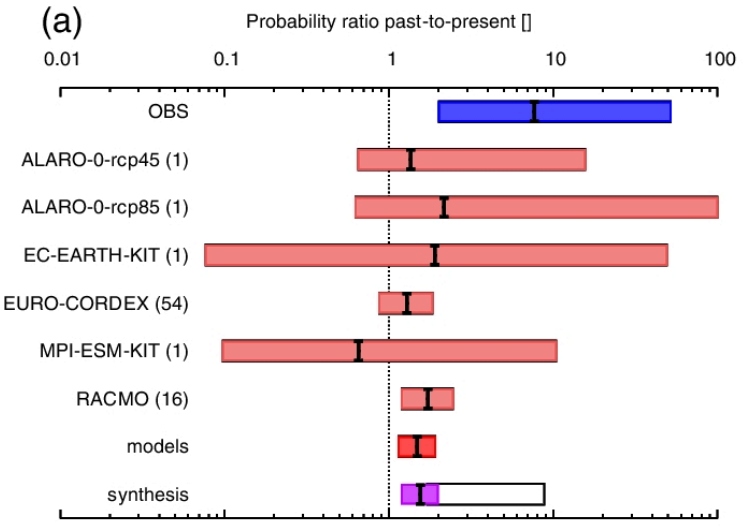
③ **Critique**: the model over-predicts after reopening: no second rise came.

④ **Repeat**: add the change in behaviour to the contact model, and refit.

Credits: Coletti et al., 2021 (CC BY 4.0).

Floods in the Vesdre valley





① **Build**: a statistical model of extreme rainfall, and climate models of two worlds.

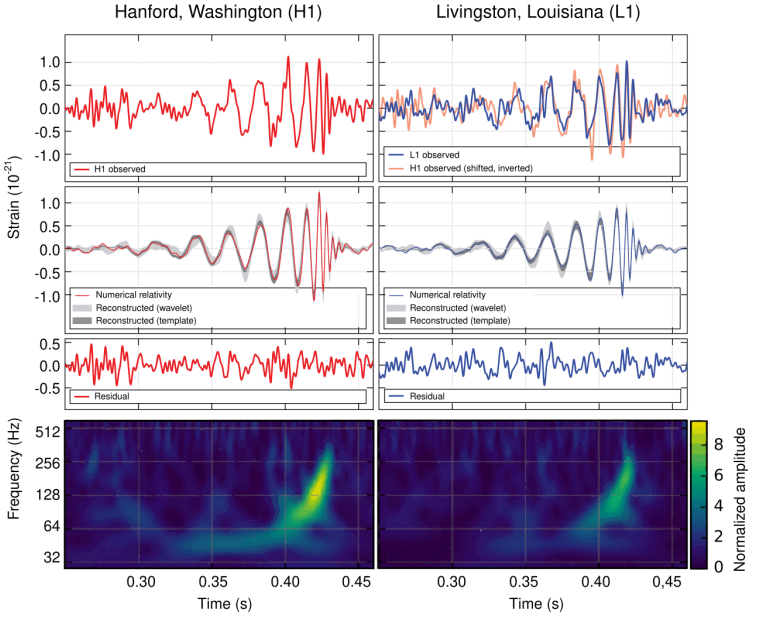
② **Compute**: the probability in each world: 1.2 to 9 times likelier today (right).

③ **Critique**: some climate models do not reproduce the observed rainfall.

④ **Repeat**: discard those models and redo the analysis, published in 2023.

Credits: Christophe Licoppe, European Commission (CC BY 4.0); Tradowsky et al., 2023, Fig. 8a, cropped (CC BY 4.0).

Gravitational wave detection



① **Build**: a waveform model of two merging black holes, from general relativity.

② **Compute**: fit the waveforms to the data, and infer the masses and distance.

③ **Critique**: no detection in eight years: the detectors were too insensitive.

④ **Repeat**: rebuild the detectors and the noise model; detection in 2015.

Credits: Abbott et al., 2016 (CC BY 3.0).



Box's loop

> *All models are wrong, but some are useful.*
>
> — George Box



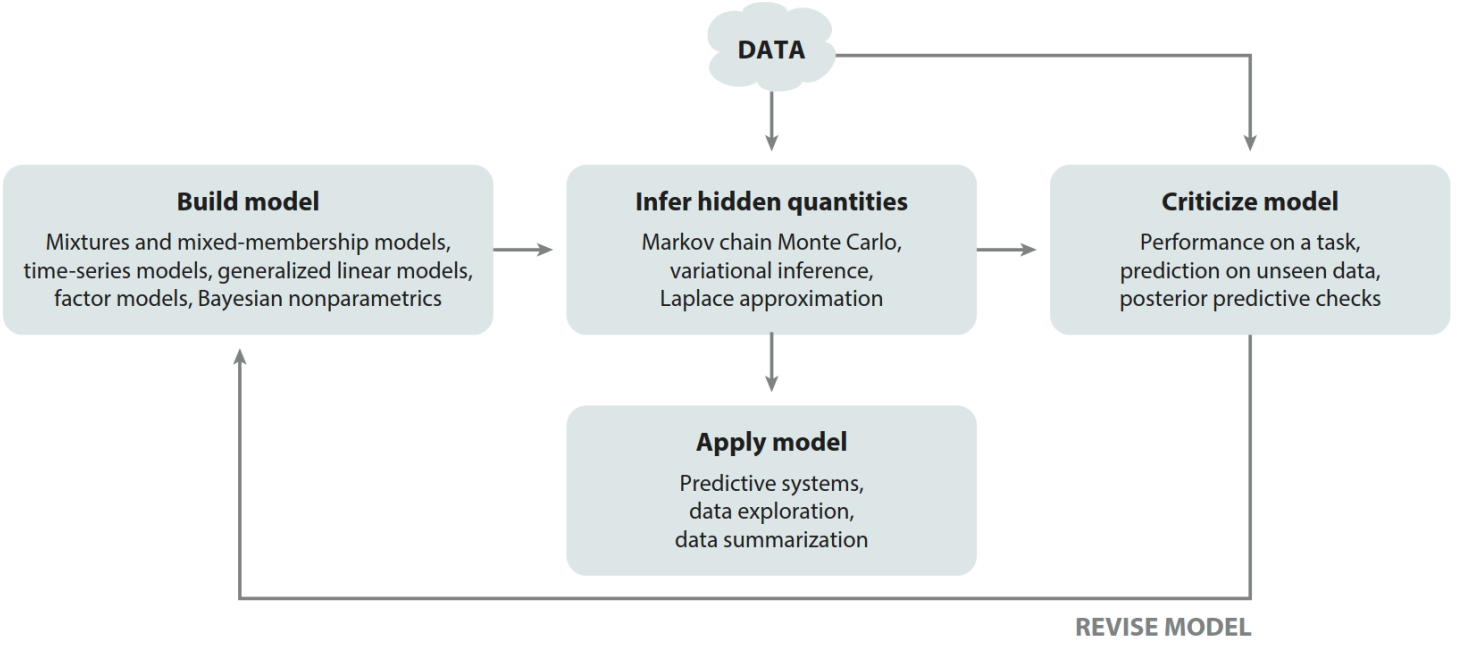
What is data science?

Data science is the discipline of **extracting knowledge** from data through the iterative application of the **scientific method**.

But how?

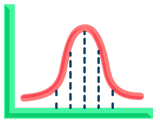


Box's loop



Scientific inquiry as an iterative process: build, compute, critique, repeat.

Credits: Blei, 2014.



Step 1: Build

The first step to understanding a phenomenon is to **build a model** of it, as a simplified representation that captures its essential aspects.

- A model specifies assumptions about the data generating process.
- It encodes domain knowledge and constraints.
- It is formulated within an appropriate mathematical abstraction.
- It defines what you observe and what you do not observe but assume exists.



Step 2: Compute

The next step is to **compute** what the data tells you about the phenomenon of interest under the assumptions of your model.

- Fitting the model to data involves solving an optimization problem.
- Inference runs the model backward, from observed data to the unobserved quantities that could have produced them.
- Prediction is used to answer questions about future or unseen data.



Step 3: Critique

The third step is to **critique** the model and its predictions, to assess whether they are consistent with the data and domain knowledge.

- Compare predictions to observed data.
- Identify model limitations and mismatches.
- Reject the model if it fails to capture key aspects of the data.

> *Whenever you are fitting a model, we recommend constructing a scenario where it will fail.*
>
> — Gelman et al.

Credits: Gelman, Vehtari, McElreath et al., Bayesian Workflow, 2026 (Figure 4.16).

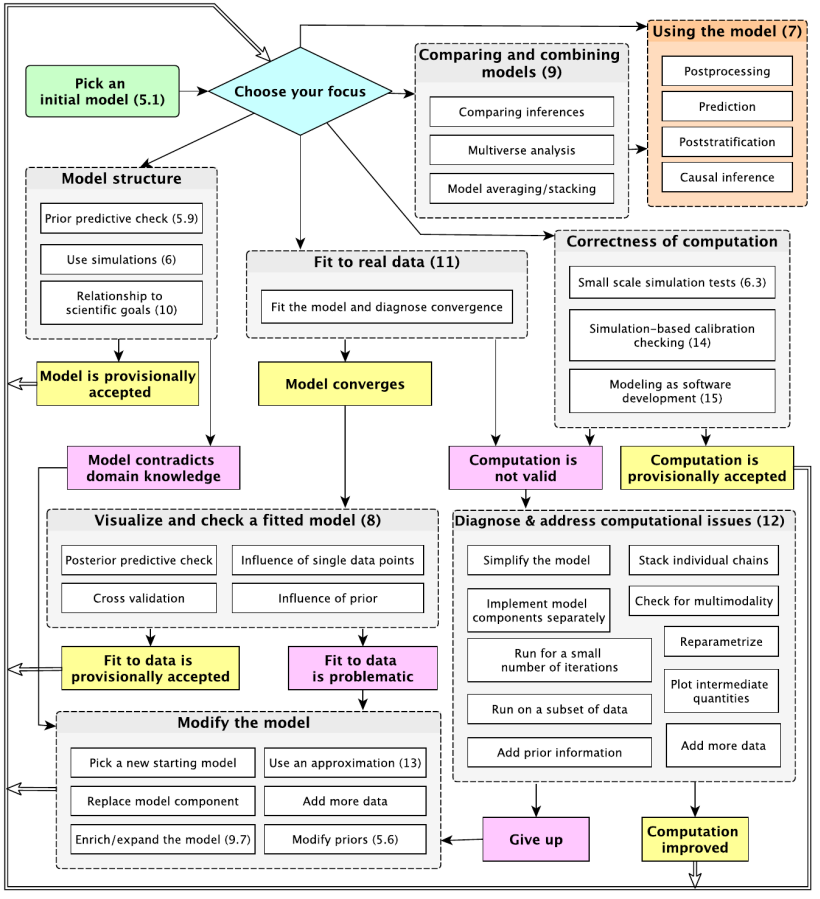


Step 4: Repeat

What you learn from the critique step informs how to **repeat** the process.

- Add complexity to the model to address its shortcomings.
- Simplify the model to improve interpretability.
- Change the model to explore alternative hypotheses.

The loop, in practice



Bayesian workflow: the steps and paths an analysis may go through.

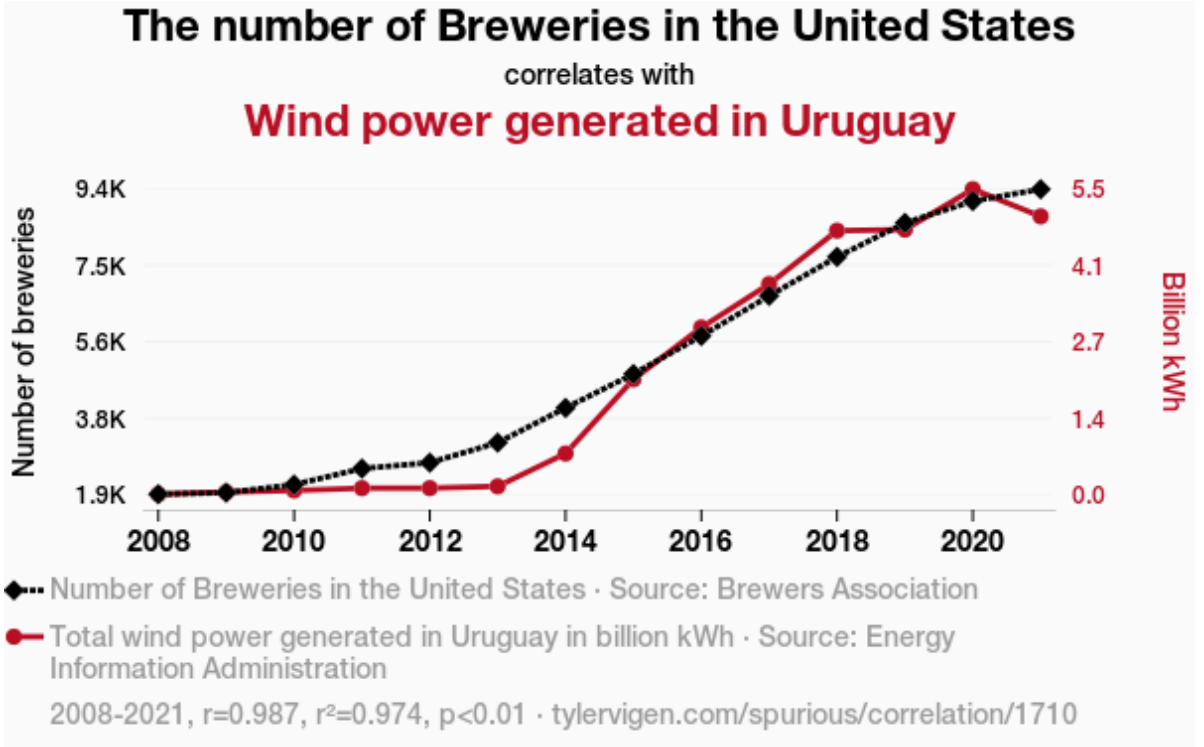
Credits: Gelman, Vehtari, McElreath et al., Bayesian Workflow, 2026 (Figure 2.1).

Why this approach matters?

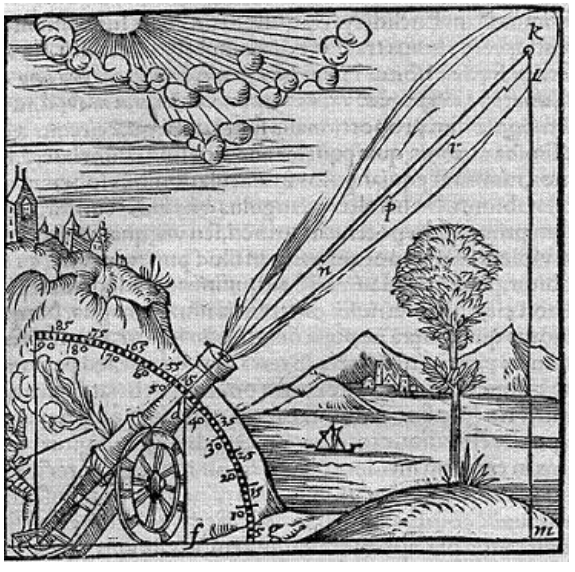
The Fourth paradigm (Hey et al, 2009) of science emphasizes the importance of data-intensive scientific discovery. However, data alone is not enough.

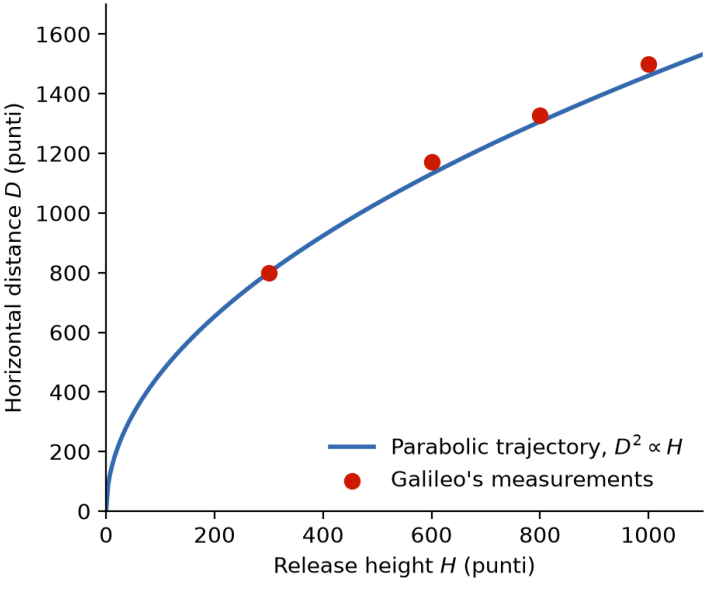
- Data without theory leads to spurious correlations.
- Theory without data leads to ungrounded speculation.
- Together, they enable **robust scientific inquiry**.

> *It is frequently assumed that big data can alleviate the need for careful modeling. We do not believe this is the case.*
>
> — Gelman et al.



Spurious correlation can easily mislead data analysis.





Theory without data: the trajectory implied by Aristotelian physics (left), against Galileo's measurements, consistent with a parabola (right).

Credits: Santbech, 1561; Galileo's folio 116v data from Breiland, 2022.

The scientific method, as embodied in Box's loop, provides a principled framework for data analysis. It contrasts with **ad-hoc data analysis practices that often lead to unreliable results**.

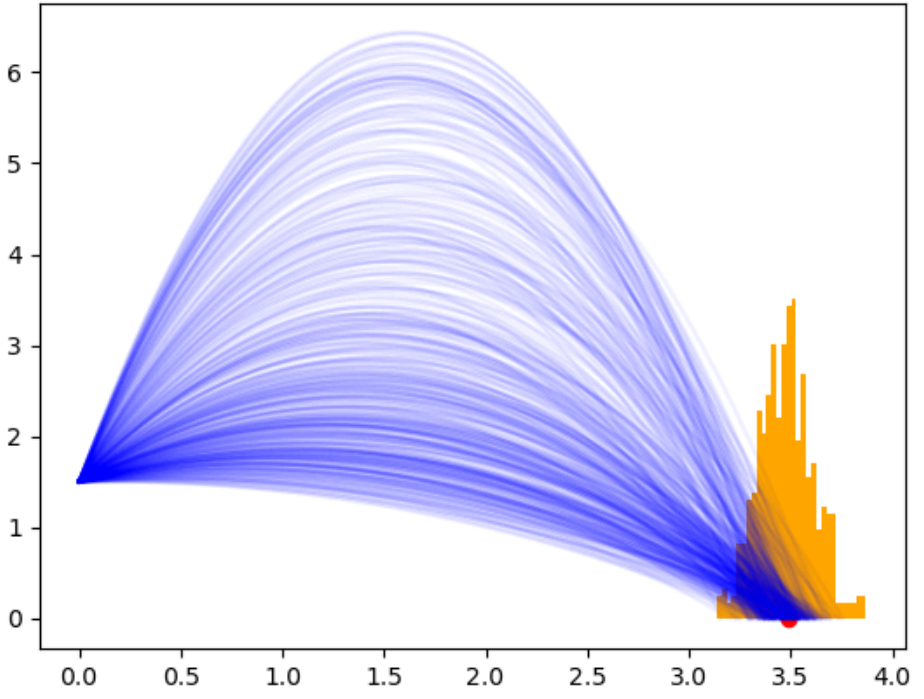
- Throw algorithms at data and see what sticks.
- Focus only on prediction accuracy.
- Treat models as black boxes.
- Stop at the first somewhat satisfactory result.

Box's loop encourages

- transparent reasoning about assumptions,
- understanding mechanisms behind data, not just correlations,
- honest assessment of model limitations,
- continuous improvement and learning.

Today's example: Projectile motion

A ball is thrown and lands at some measured distance x. What can we infer about the initial velocity v and angle α of the throw?



Let's code!

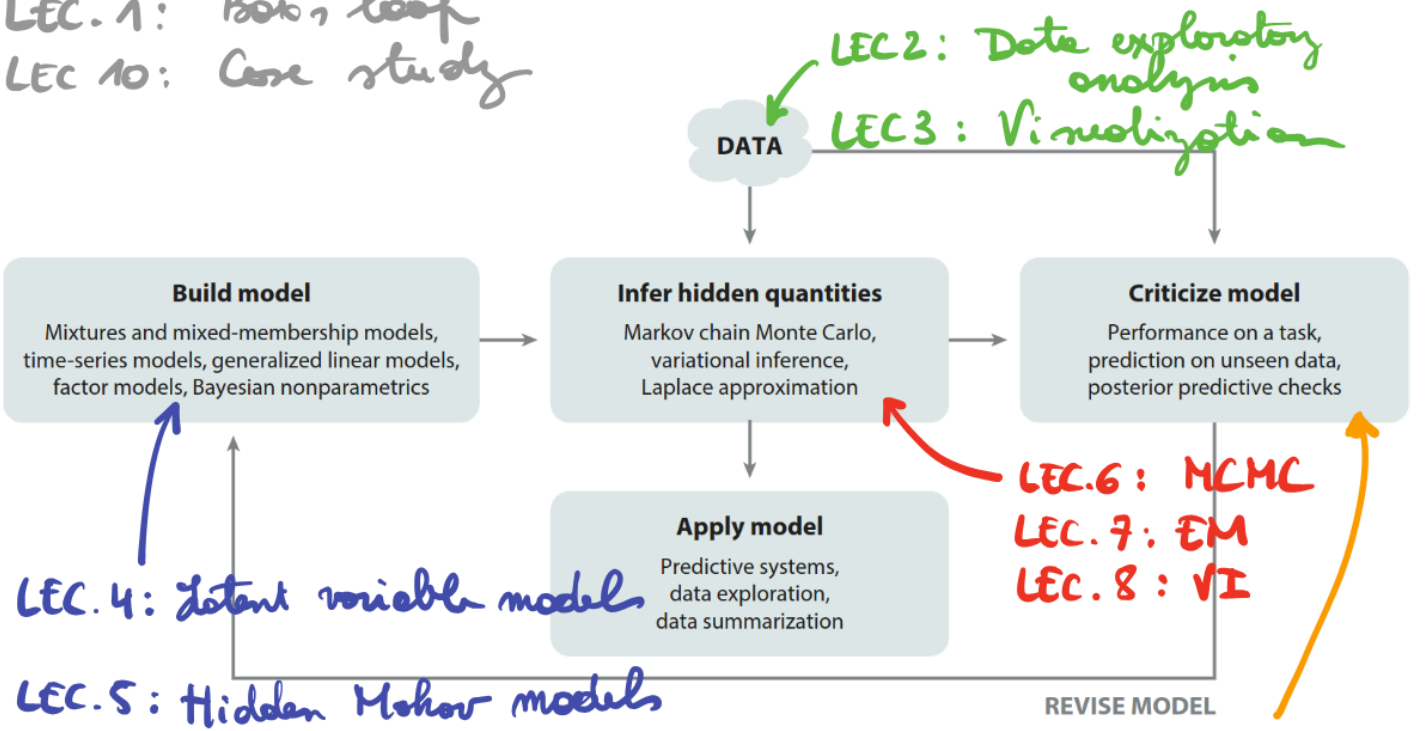
- Open a Jupyter notebook and follow along (or check `nb01.ipynb` after the lecture).
- Implement Box's loop step by step:
    1. Build a simple physical model of projectile motion.
    2. Compute estimates of v and α from data.
    3. Critique the model fit and predictions.
    4. Repeat by refining the model to account for more realistic factors.

DATS0001

Outline

- Lecture 1: Build, compute, critique, repeat
- Lecture 2: Data and exploratory analysis
- Lecture 3: Visualization
- Lecture 4: Latent variable models
- Lecture 5: State-space models
- Lecture 6: Markov chain Monte Carlo
- Lecture 7: Model criticism and validation
- Lecture 8: Expectation-Maximization
- Lecture 9: Variational Inference
- Lecture 10: Simulation-based inference
- Lecture 11: Case study

LEC. 1: Box's loop
LEC 10: Case study



My mission

Teach you how to think like a scientist in the age of data.

By the end of this course, you will be able to explore, model, and reason about data in a principled way, but also to communicate your findings effectively.

The end.